

Experiment 1

Internal Environment Experiment

Objective

The purpose of this experiment was to investigate whether the choice of drying material influences the internal environment experienced by stored maize. Humidity, carbon dioxide concentration, and temperature were monitored within sealed bags constructed from polyvinyl chloride (PVC), polypropylene (PP), and nylon. These variables were selected as proxies of conditions that may influence fungal growth and consequently the risk of aflatoxin contamination.

Motivation

Aflatoxin contamination is strongly associated with poor drying and storage conditions, particularly elevated moisture levels. Different plastics possess different permeabilities to water vapour and gases such as oxygen and carbon dioxide meaning that material selection may affect the microenvironment surrounding crops in the drying container.

The aim of this experiment was therefore to determine whether commonly available plastic materials produce measurable differences in internal environment in terms of humidity, temperature, and carbon dioxide concentration during storage. If significant differences exist, material selection may represent a simple means of improving storage performance within the SolarSafe system.

Materials

- Polyvinyl chloride (PVC) bags
- Polypropylene (PP) bags
- Nylon bags
- Corn
- Carbon dioxide sensor
- humidity sensor
- Temperature sensor
- Humidity Probe

Experimental Setup

Approximately equal masses of corn (190g) were placed into bags constructed from PVC, PP, and nylon. Sensors capable of measuring relative humidity, carbon dioxide concentration, and temperature were positioned within each bag. Bags were then sealed (with tape) and monitored over the duration of the experiment (around 5 days). The corn were initially in the fridge prior to being placed in bags to maintain uniform conditions as much as possible.

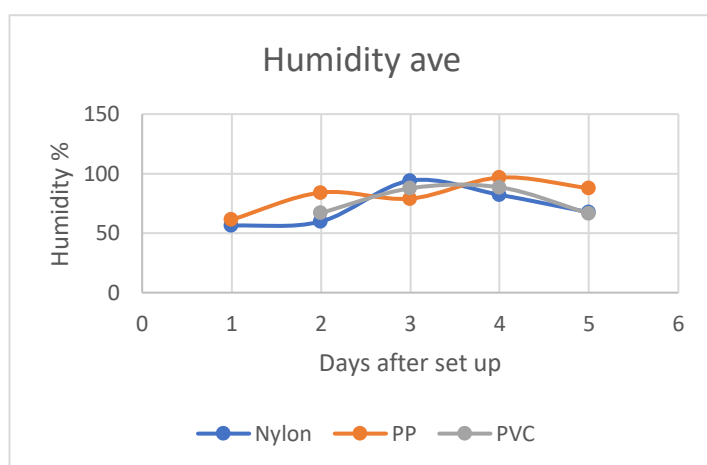
Figure 1: Nylon bag with corns 5 and 6 and controls left to dry outside 7 and 8.



Method

- 1) Initial maize moisture content was measured using a moisture probe.
- 2) Sensor readings for relative humidity, carbon dioxide concentration, and temperature were recorded daily over the five-day experimental period.
- 3) Final maize moisture content was measured at the end of the experiment.
- 4) Environmental conditions recorded in the different bag materials were compared.

Though initially we had set out to measure Oxygen permeability, after realising CO₂ sensors were readily available in the Materials Lab these were employed instead. The plastic bag systems were maintained as sealed as possible until Monday (approximately Day 3). At this point, the bags were opened to simulate the periodic opening and closing of practical drying containers and to obtain sensor readings, as several CO₂ sensors saturated at 9999 ppm and meaningful measurements couldn't be inferred at this limit.



Relative Humidity

Relative humidity increased substantially in all systems.

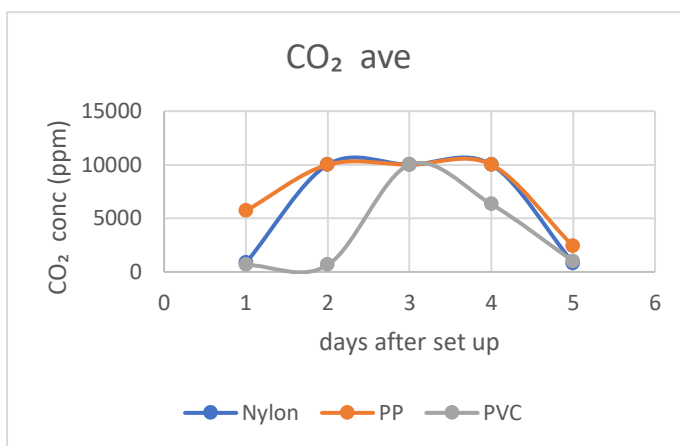
- PP increased from 61.3% to a maximum of 96.7%.
- Nylon increased from 56.2% to 94.1%.
- PVC increased from 66.8% to 88.5%.

Following the opening of the bags on Day 3, humidity decreased in all treatments:

- Nylon: 82.2% → 67.5%
- PVC: 88.5% → 66.5%
- PP: 96.7% → 87.8%

The increase in humidity indicates that moisture released from the corn accumulated within the sealed bags. This is expected because corn continues to respire releasing water. For drying applications, high humidity is undesirable because it reduces the moisture gradient between the corn and the surrounding environment, slowing further drying. PP consistently exhibited the highest humidity, suggesting that it was the most effective barrier to water vapour transfer trapping moisture. PVC maintained lower humidity levels than PP and therefore appears more favourable for drying applications. Nylon had intermediate values.

Carbon Dioxide (CO₂)



CO₂ concentrations rose throughout the experiment in all bag materials. The nylon and PP treatments reached the sensor maximum reading of 9999 ppm within 24 hours, whereas the PVC treatment exhibited a slower rate of CO₂ accumulation and reached sensor saturation only at a later time point. After the bags were opened, CO₂ concentrations declined markedly. However, concentrations remained highest in the PP treatment (2396 ppm), compared with PVC (994 ppm) and nylon (834 ppm).

The increase in CO₂ confirms that the corn remained biologically active throughout storage.

Corn respiration can be described by:

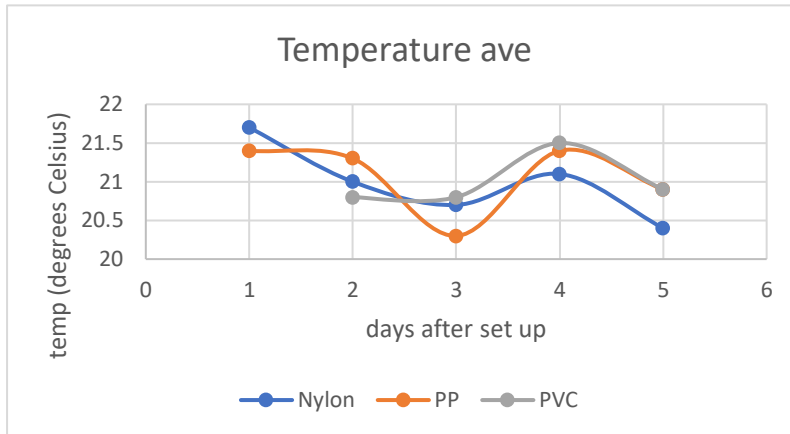


The accumulation of CO₂ therefore provides evidence of ongoing respiration within the stored grain. PP consistently retained the highest CO₂ concentrations, suggesting reduced gas exchange and ventilation. PVC exhibited lower CO₂ accumulation, indicating greater gas permeability.

However, several readings reached the sensor maximum value of 9999 ppm. These values indicate that the true concentration exceeded the sensor measurement range and therefore represent minimum estimates rather than exact concentrations. Alongside this there are few

data points to draw trends from and these may be susceptible to sensor noise/drift over the course of the experiment.

Temperature



Temperature remained relatively stable throughout the experiment, varying from approximately 20.3°C to 21.7°C, a total range of less than 1.5°C (Figure X). No substantial differences in temperature were observed between the packaging treatments despite marked increases in CO₂ concentration.

As respiration generates both heat and carbon dioxide, we anticipated perhaps higher rates of respiration might result in increased temperatures. However, the limited temperature variation observed suggests that respiratory heat production was either relatively small or effectively dissipated to the surrounding environment. Consequently, temperature is unlikely to have been a major factor contributing to differences between treatments. Instead, the observed differences in humidity and moisture content are more likely to reflect differences in the barrier properties of the packaging materials.

The use of additional temperature sensors at multiple locations within each storage system may have provided a more detailed understanding of the spatial distribution of heat generated by respiration and its dissipation to the surrounding environment. The same is the case for humidity and CO₂ measurements too.

Moisture Distribution Within Individual Corn Cobs

To investigate moisture distribution within individual cobs, moisture measurements were collected at four locations (North, East, South and West) using a two-prong moisture probe. Radar plots were generated for representative cobs from each treatment (Corn 2, Corn 4, Corn 6 and Corn 8) to visualise moisture gradients and identify potential localised hotspots.

Initial Moisture Distribution

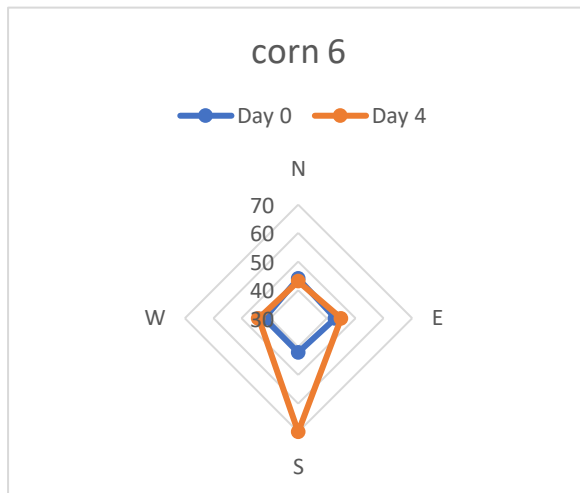
At Day 0, moisture content was relatively uniform across all cobs, with most measurements ranging between 42–44%. This suggests that moisture was initially distributed

homogeneously throughout the corn and that no major moisture gradients existed prior to storage.

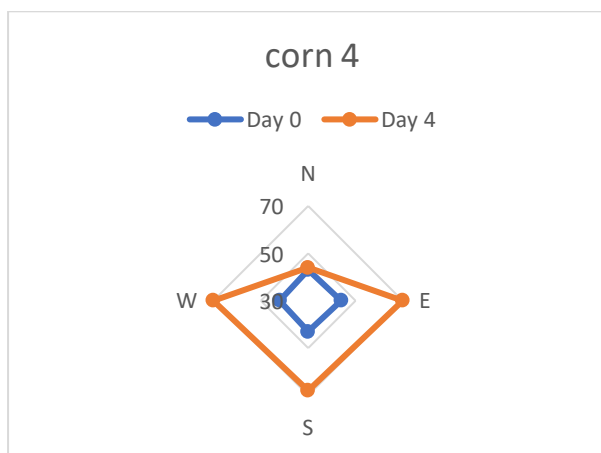
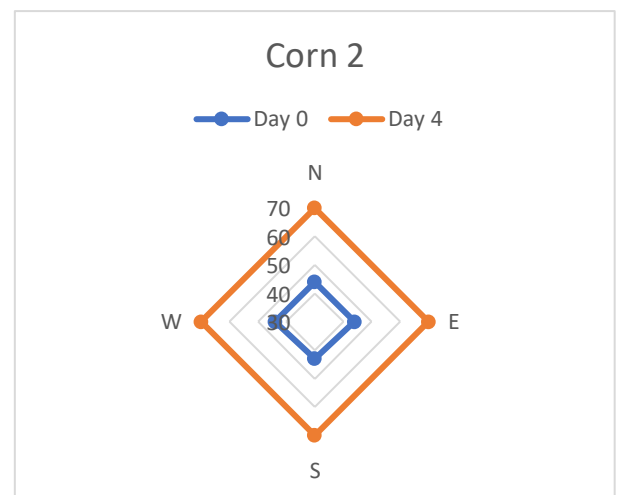
Day 4 Moisture Distribution

By Day 4, differences in moisture distribution had developed between treatments. Radon Graphs of moisture distribution throughout corns:

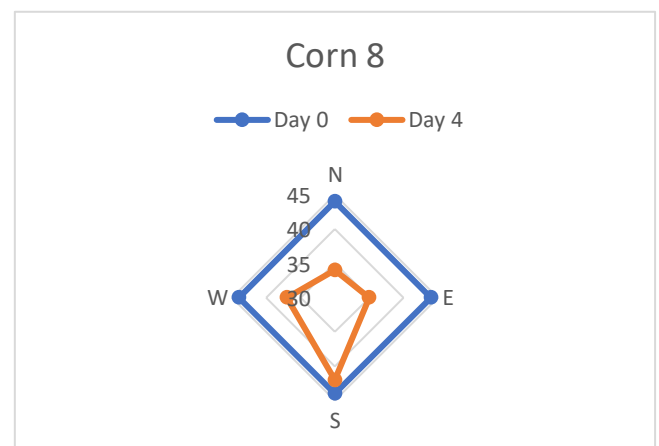
In Nylon corn 6



In PVC corn 2



PP corn 4



Control (outside)

control (not in plastic bag) corn 8

Corn 2 (PVC) exhibited a relatively uniform increase in moisture, with all measured positions reaching approximately 70%. This suggests that moisture accumulation occurred evenly throughout the cob and that the internal environment promoted a relatively homogeneous moisture distribution. Corn 4 (PP) displayed substantial spatial variability. Moisture contents ranged from approximately 44% to 70%, indicating the development of localised regions of elevated moisture. Corn 6 (Nylon) showed a similar pattern, with moisture contents ranging from approximately 43% to 70%. These large gradients suggest that moisture redistribution within some cobs was not uniform. The control sample (Corn 8) remained substantially drier overall, with moisture contents ranging from approximately 34-42%. Although some

variation was present, the magnitude of the gradient was considerably smaller than that observed in Corn 4 and Corn 6.

Moisture distributions varied both between drying systems and between individual cobs within the same system, ranging from relatively homogeneous to highly heterogeneous patterns. The occurrence of both heterogeneous and homogeneous moisture distributions within the same packaging treatments suggests that moisture redistribution was influenced by factors beyond the packaging material alone. Possible contributors include cob position within the package, local airflow, condensation, contact with surfaces, and natural variation between cobs.

These localised moisture hotspots may be important from a drying/storage perspective, as fungal growth is often initiated in isolated regions of elevated moisture rather than uniformly throughout the grain mass. Consequently, average moisture content alone may not fully reflect storage performance or aflatoxin risk. However, moisture measurements were obtained using a handheld two-prong probe at only four locations per cob, limiting spatial resolution. Although smaller gradients may have gone undetected, the results demonstrate that localised moisture accumulation can occur within stored maize cobs. *

Conclusion

Packaging material influenced the storage environment experienced by the maize. Polypropylene (PP) consistently produced the highest relative humidity and resulted in the highest final maize moisture content, indicating the least favourable conditions for drying. PVC maintained lower humidity levels and resulted in the lowest final maize moisture content of the enclosed systems, while nylon showed intermediate moisture retention.

Carbon dioxide concentrations increased substantially in all treatments, confirming continued respiration of the stored maize. Following bag opening, nylon exhibited the lowest measured CO₂ concentration (834 ppm), followed by PVC (994 ppm), while PP remained highest (2396 ppm). Temperature remained similar across all treatments and therefore appeared to have little influence on storage performance.

Overall, PVC provided the most favourable conditions for moisture removal, PP the least favourable, and nylon exhibited intermediate drying performance while also showing the lowest measured post-opening CO₂ concentration. Although humidity, moisture content and CO₂ concentration are useful indicators of storage conditions, they represent indirect measures of aflatoxin risk. Future studies should therefore focus on establishing direct links between packaging material, fungal growth, and aflatoxin production under realistic storage conditions.

Sodium Hypochlorite Durability Testing

Although SolarSafe primarily aims to reduce aflatoxin contamination through improved drying, fungal contamination may also arise from contact with contaminated soil, surfaces, or airborne spores. Sodium hypochlorite is a low-cost disinfectant that is widely available in Kenya and could provide a practical method of reducing external microbial contamination. However, before such a strategy can be implemented, it is necessary to determine whether repeated exposure to sodium hypochlorite causes degradation of candidate container materials. This experiment therefore assessed the resistance of PVC, polypropylene (PP), and nylon to repeated sodium hypochlorite exposure.

Materials

- PVC samples
- Polypropylene (PP) samples
- Nylon samples
- Sodium hypochlorite solution (3%)
- Sodium hypochlorite solution (5%)
- Distilled water (control)
- Laboratory oven
- ATR spectroscopy equipment
- Microscope (optical)

Experimental set up

Sample preparation, labelling, chemical exposure, and ageing studies were carried out in the Plastics Laboratory. PVC, polypropylene (PP), and nylon samples were cut into squares approximately 1.5 cm × 1.5 cm in size. Three replicate samples were prepared for each material and treatment condition. Because the samples were flexible, those intended for ATR-FTIR analysis were attached to glass microscope slides using double-sided tape to make handling and measurement easier.

The following treatment groups were established; all solutions made up to 30ml:

Treatment	Purpose
Water control	Baseline comparison
3% sodium hypochlorite	Representative disinfectant exposure
5% sodium hypochlorite	Elevated chemical exposure
5% sodium hypochlorite + elevated temperature	Accelerated ageing condition

Following exposure some samples were transferred to an oven to simulate repeated use and long-term environmental exposure.

Experiment and Sample



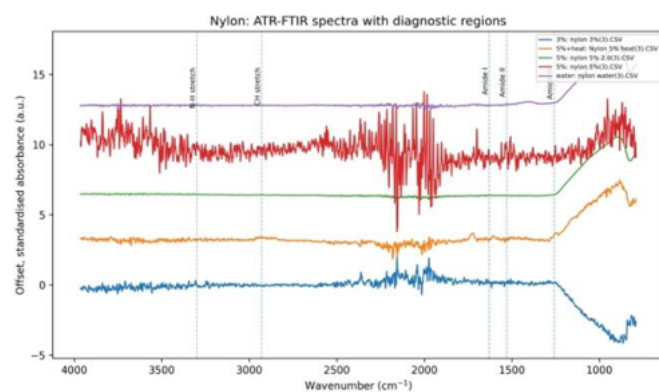
set up:

ATR samples on glass slides.

Material degradation was assessed using two complementary techniques:

ATR-FTIR Analysis

Attenuated Total Reflectance Fourier Transform Infrared Spectroscopy (ATR-FTIR) was used to investigate chemical and structural changes in nylon, PVC and polypropylene (PP) following treatment. ATR-FTIR measures how infrared light is absorbed by a material. Different chemical bonds absorb and resonate at infrared radiation at their characteristic frequencies, producing a spectrum that acts as a molecular fingerprint of the polymer. Changes in peak position, intensity or shape can indicate chemical degradation, oxidation, additive loss, changes in molecular organisation or alterations in crystallinity. For each material, spectra from treated samples were compared with the corresponding water-treated control. Offset spectra are shown in the graphs below to allow visual comparison between treatments. The background was collected first to dispense of its effects in the samples.



Nylon

The offset ATR spectra for nylon.

The nylon spectra contained characteristic amide bands at approximately 1630 cm^{-1} (Amide I), 1530 cm^{-1} (Amide II) and 1260 cm^{-1} (Amide III), which arise from the amide groups forming the polymer backbone. Despite evident noise in the data, the heat-treated sample showed

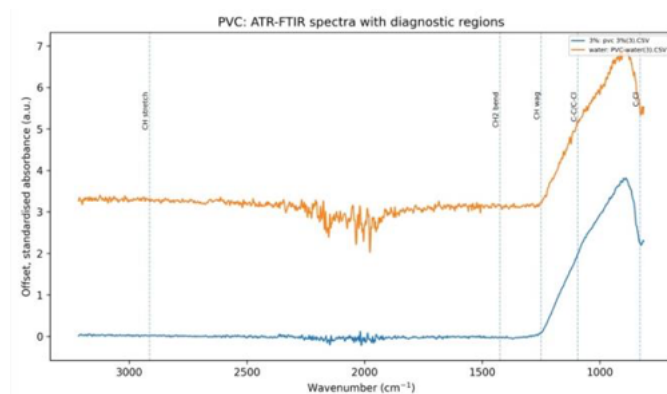
the largest deviation from the water-treated control, although the characteristic amide peaks

remained present. This indicates that the nylon backbone remained largely intact while the local molecular environment surrounding the amide groups was altered.

We have decided not to incorporate images from microscopy into the reports due to ambiguity and the fact that there are 11 samples. However we did note observations such as for nylon microscopy revealed fibre curling, contraction and enhanced longitudinal grooves following heat treatment. These changes indicate that the fibre morphology was altered substantially relative to the control. Because nylon is manufactured under tension, heating can relieve residual internal stresses and allow partial rearrangement of polymer chains into a lower energy configuration. The observed curling and contraction are consistent with stress relaxation of the oriented fibres and may reflect local compaction of the fibre structure.

The agreement between ATR and microscopy suggests that heat treatment primarily induced structural reorganisation and chain relaxation rather than severe chemical degradation. Among all treatments, thermal exposure produced the most pronounced changes in nylon, indicating that heat had a greater effect on the material than sodium hypochlorite exposure alone.

PVC



The offset ATR spectra for PVC.

PVC spectra are dominated by the fingerprint region between approximately 1250 and 800 cm^{-1} , which contains vibrations associated with the polymer backbone and C-Cl bonds. This region is particularly

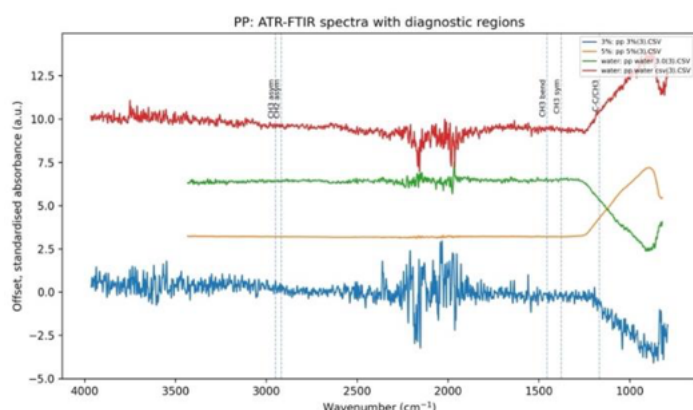
sensitive to changes in additives, molecular packing and local polymer structure.

Interpretation of the PVC spectra was complicated by substantial experimental noise, particularly in the 5% sodium hypochlorite and heat-treated samples. Although some differences relative to the control were apparent, the ATR data alone do not provide conclusive evidence for extensive chemical degradation.

Microscopy provided clearer evidence of treatment effects. Compared with the control, treated PVC samples exhibited increased roughness, localised pitting and heterogeneous surface texture. One possible explanation is partial extraction or redistribution of plasticisers and other additives within the PVC matrix. Plasticiser loss can alter surface morphology without necessarily causing significant degradation of the polymer backbone. Another possible mechanism is early stage dehydrochlorination, the classical degradation pathway of PVC, although the present ATR data do not provide sufficient evidence to confirm this process.

Taken together, the results indicate that sodium hypochlorite treatment modified the PVC surface. Again the observations are most consistent with surface restructuring and possible

additive loss rather than catastrophic degradation of the material.



Polypropylene (PP)

The offset ATR spectra for PP.

Polypropylene exhibited the smallest treatment-dependent spectral changes of the three polymers. Characteristic bands associated with CH stretching and CH₃ bending remained largely unchanged across

all treatments, indicating that the molecular structure of the remaining polymer was largely preserved.

Microscopy similarly revealed only moderate surface roughening and did not show the extensive pitting observed in PVC or the fibre restructuring observed in nylon. These observations suggest that PP experienced relatively limited chemical modification at the surface. However, direct observations during treatment revealed softening and partial dissolution of PP under the heated 5% sodium hypochlorite condition. This behaviour was not strongly reflected in either the ATR spectra or microscopy images (especially considering the fact we could not make a sample for PP under heated conditions), and highlights an important distinction between chemical stability and functional stability. Overall, PP appeared chemically resistant but exhibited reduced physical integrity under the most aggressive treatment conditions. The results therefore suggest limited chemical degradation but potentially significant loss of practical performance.

Potential microscopy analysis

Quantitative image analysis was considered as a means of supplementing visual microscopy observations. One potential approach would be to extract multiple texture descriptors from microscopy images and apply Principal Component Analysis (PCA) to reduce these variables into a small number of components capturing the major sources of variation between samples. This would allow samples with similar surface morphologies to be grouped and treatment-induced changes to be quantified objectively. However, due to time constraints and the limited number of images available, a robust PCA analysis was not performed in the present study. Future work incorporating a larger image dataset could provide a more rigorous quantitative assessment of surface degradation and morphological change.

Comparison of Materials

The microscopy and ATR-FTIR results were generally in agreement. Nylon was affected mainly by heat treatment, showing evidence of structural reorganisation but limited chemical degradation. PVC exhibited the greatest degree of surface modification, with increased roughness and pitting observed after treatment. Polypropylene showed comparatively small spectroscopic and microscopic changes, although samples exposed to heated 5% sodium hypochlorite softened and partially dissolved.

Overall, PVC appeared most susceptible to surface modification, nylon primarily underwent thermal restructuring, and polypropylene exhibited the greatest reduction in physical stability under severe treatment conditions.

Profilometry

Profilometry was intended to assess changes in surface morphology and roughness following treatment. In this technique, a probe is scanned across the material surface to generate a surface profile, allowing quantitative measurements of roughness and other topographical features. Increases in surface roughness can be indicative of material degradation and may affect long-term durability and cleanability.

However much to everyone's disappointment, profilometry measurements could not be performed because the instrument was unavailable due to a technical fault at the time of testing. Consequently assessment of surface degradation was limited to visual observations and ATR-FTIR analysis.

Limitations

Several limitations should be considered when interpreting the results of this experiment:

- The study investigated only three candidate materials (PVC, PP, and nylon). Other commercially available polymers may demonstrate superior resistance to sodium hypochlorite exposure.
- Accelerated ageing conditions may not fully replicate the environmental conditions experienced by SolarSafe containers during long-term deployment in Kenya. Factors such as UV exposure, rainfall, and mechanical wear were not investigated.
- ATR spectroscopy and profilometry primarily characterise surface changes. Whilst these techniques can identify evidence of chemical degradation and changes in surface morphology, they do not directly measure changes in bulk mechanical properties such as tensile strength or flexibility.
- The experiment assessed material durability rather than antimicrobial efficacy. Although sodium hypochlorite is widely used as a disinfectant, the ability of the treatment to reduce fungal contamination or aflatoxin levels was not directly measured.
- Resource and time constraints limited the duration of testing and prevented longer-term ageing studies.

Recommendations for Future Work

- Measure changes in tensile strength and flexibility following chemical exposure.
- Investigate the effects of UV ageing alongside sodium hypochlorite treatment.
- Test additional candidate materials.
- Quantify the antimicrobial effectiveness of sodium hypochlorite under realistic operating conditions.
- Directly investigate whether external contamination contributes significantly to aflatoxin contamination within the SolarSafe drying system.

EXPERIMENT 3

Experiment: Aflatoxin B1 Binding Affinity to Food-Contact Plastics

Motivation and Significance

The previous experiments assessed plastic performance from two complementary perspectives:

Previous experiment	What it tested
Internal condition monitoring	How plastics influence drying conditions such as humidity, moisture retention and CO ₂ accumulation
Hypochlorite durability testing	How plastics withstand repeated chemical cleaning and environmental stress
Proposed aflatoxin-binding assay	How plastic surfaces interact directly with aflatoxin molecules

Together, the previous experiments established how the materials behave under storage and cleaning conditions. However, neither experiment addressed a more fundamental question: how do the plastics interact directly with contaminants encountered during food storage?

The proposed aflatoxin-binding assay therefore examines a different aspect of material performance. Rather than focusing on environmental control, it investigates whether the polymer surface itself influences the behaviour of aflatoxin molecules. The key question is not simply: Which plastic prevents aflatoxin contamination best? But rather, does the plastic itself bind aflatoxin, and could this contribute to toxin accumulation or retention?

This question is motivated by observations from the packaging literature. Several studies have reported lower aflatoxin concentrations in foods stored in PET packaging compared with polyethylene-based materials. For example, Macri et al. found that nuts and dried fruits packaged in PET exhibited lower aflatoxin B1 contamination than equivalent products packaged in LDPE. Similarly, pistachio storage studies reported that increased moisture uptake was associated with greater aflatoxin accumulation and that most materials, with the exception of LDPE delayed this process.

However, these studies do not fully explain why PET performs better than other materials. At least two mechanisms may contribute:

Mechanism	Explanation
Barrier effect	PET may limit moisture and oxygen transfer, reducing fungal growth and subsequent aflatoxin production
Surface interaction effect	PET may interact differently with aflatoxin molecules, resulting in reduced adsorption or retention at the polymer surface

Most published storage studies primarily evaluate the barrier properties of packaging materials. The proposed assay instead isolates surface interactions by exposing plastic samples directly to a known aflatoxin B1 solution in the absence of food, moisture-driven fungal growth, and other storage/drying related variables. This allows toxin-polymer interactions to be investigated independently of drying/storage conditions.

Aim

To compare the adsorption of aflatoxin B1 onto different food-contact plastics.

Hypothesis

Different plastics will adsorb different amounts of aflatoxin B1 due to differences in surface chemistry, hydrophobicity, crystallinity, and polymer structure.

Based on observations from the packaging literature, the following trend is anticipated:

Plastic	Expected performance
PET	Lowest aflatoxin adsorption / likely best
HDPE / PE	Intermediate
PP	Intermediate
LDPE	Highest aflatoxin adsorption / likely worst
PVC	Unknown; included because it is currently used within the SolarSafe system

Experimental Design

Samples of PET, PP, HDPE, LDPE and PVC will be cut to identical dimensions and cleaned using a standardised washing procedure. Glass samples will be included as a low-binding control, together with no plastic controls to account for any natural aflatoxin degradation during incubation.

Each sample will then be immersed in a known concentration of aflatoxin B1 solution under controlled temperature, pH and lighting conditions. By maintaining identical conditions for all materials, any differences in aflatoxin concentration can be attributed primarily to interactions with the polymer surface.

Aflatoxin Quantification

Following incubation, the concentration of aflatoxin B1 remaining in solution will be measured. The principle of the assay is that adsorption of aflatoxin onto the plastic surface will reduce the concentration remaining in the surrounding solution. Plastics that bind greater amounts of aflatoxin are therefore expected to produce lower final solution concentrations.

Samples will be collected at several time points (for example 0, 1, 4 and 24 hours) to assess both the rate and extent of adsorption. Measuring concentration over time allows adsorption kinetics to be investigated and indicates whether adsorption reaches a stable equilibrium value.

Two analytical approaches could be used.

ELISA provides a rapid and relatively inexpensive screening method suitable for analysing large numbers of samples. The assay uses antibodies that recognise aflatoxin B1 and provides an estimate of the concentration remaining in solution.

HPLC provides greater specificity and quantitative accuracy by separating aflatoxin from other compounds before measurement. Although more resource-intensive, it is widely regarded as a reference method for aflatoxin analysis.

A practical approach would be to use ELISA for initial screening and HPLC to confirm any significant differences observed between materials.

Data Analysis

Adsorption will be calculated using:

$$\text{Binding (\%)} = ((C_0 - C_t)/C_0) \times 100$$

where C_0 is the initial aflatoxin concentration and C_t is the concentration remaining in solution after incubation.

Higher binding values indicate greater adsorption of aflatoxin onto the plastic surface, whereas lower values indicate weaker interaction with the toxin.

Desorption Assay

Adsorption measurements alone do not indicate whether aflatoxin is retained permanently by the polymer surface. To investigate this, a secondary desorption experiment may be performed.

Following adsorption, the plastic coupons will be removed from the aflatoxin solution, gently rinsed, and transferred into fresh aflatoxin-free buffer. After a second incubation period, the concentration of aflatoxin released into the clean solution will be measured.

This experiment distinguishes between strong and weak surface interactions. A plastic that adsorbs aflatoxin but releases very little into fresh buffer would be considered a strong binder. In contrast, a plastic that readily releases adsorbed aflatoxin may act as a temporary reservoir, storing toxin on its surface before releasing it later. Measuring both adsorption and desorption therefore provides a more complete assessment of toxin–polymer interactions than adsorption measurements alone.

Expected Significance

This experiment addresses a mechanistic question not examined in the previous studies: whether polymer surfaces directly influence aflatoxin behaviour. Combined with the storage-environment and material-durability experiments, the results would provide a more complete assessment of material suitability for food-contact applications.

A material that maintains favourable drying conditions but strongly retains aflatoxin may present different risks from a material that exhibits little toxin adsorption. Understanding these interactions would therefore help separate the effects of storage environment from the intrinsic properties of the polymer surface and may explain some of the differences reported between packaging materials in the literature.

References

Macri, A.M., Pop, G., Epure, D.G., Szabo, K. and Vodnar, D.C. (2021) 'The Occurrence of Aflatoxins in Nuts and Dry Nuts Packed in Four Different Plastic Packaging from the Romanian Market', *Microorganisms*, 9(1).

Omar, S.S., Haddad, M.A. and Parisi, S. (2020) 'Validation of HPLC and Enzyme-Linked Immunosorbent Assay (ELISA) Techniques for Detection and Quantification of Aflatoxins in Different Food Samples', *Foods*, 9(5).

Shakerardekani, A. and Karim, R. (2013) 'Effect of Different Types of Plastic Packaging Films on the Moisture and Aflatoxin Contents of Pistachio Nuts During Storage', *Journal of Food Science and Technology*, 50(2), pp. 409–411.

Future Directions

The present study focused on materials that are readily available and compatible with the current SolarSafe design. However, future development should not be constrained by existing design choices. For example, flexible plastic containers may not represent the optimal solution if more rigid materials provide superior durability or contamination control. Similarly the use of a black inner surface may warrant reconsideration, as increased heat absorption could promote temperature gradients within the container and potentially create localised hotspots that favour fungal growth.

Future work should move beyond the constraints of the current design and explore alternative materials and container configurations that may improve drying performance, minimise temperature gradients, and reduce aflatoxin risk. There may also be opportunities to develop a Kenya-specific aflatoxin mitigation product, analogous to the biological control strategy employed by Aflasafe, but tailored to local storage practices and available materials. Drawing inspiration from successful technologies developed in other countries while adapting them to local agricultural and economic conditions may offer the greatest potential for long-term impact.

Overall Conclusion

This project evaluated candidate materials for the SolarSafe drying system from three perspectives: drying performance, material durability under disinfection, and aflatoxin interactions.

The internal environment experiment demonstrated that material choice significantly influenced drying conditions. PVC generally produced the most favourable drying environment, whereas PP exhibited the highest humidity, moisture retention, and CO₂ accumulation. The hypochlorite durability study showed that the materials responded differently to chemical and thermal stress, with nylon appearing to undergo more structural reorganisation, PVC exhibiting surface modification, and PP showing reduced physical stability under the most aggressive treatment conditions.

The proposed aflatoxin-binding assay extends this work by investigating whether polymer surfaces interact directly with aflatoxin B1. Unlike conventional storage studies, this experiment would isolate toxin–polymer interactions from the effects produced by internal environments.

Taken together, these studies suggest that material selection for SolarSafe should not be based solely on drying performance. A suitable material must also withstand repeated use and cleaning while minimising any interactions that could contribute to aflatoxin retention. Considering all three factors provides a more complete basis for selecting materials for future SolarSafe development.
